

HEALTHCONNECT CLINIC — FINAL ANALYTICS & DECISION SUPPORT REPORT

Data Analytics Track | AnalystLab Africa Experience Lab | Ojo Precious Mojolaoluwa | Weeks 4–8

1. Executive Summary

HealthConnect Clinic was losing almost half of its booked appointments to no-shows. Over five weeks I measured the problem, found what drives it, tested my own numbers until they held up, and turned them into three cheap, targeted actions.

The headline: of 5,000 appointments, 2,423 (48.5%) ended in a no-show, while only 263 (5.3%) were properly cancelled. Excluding cancellations — the definition I agreed with the Data Science track — the no-show rate is 51.2%.

Three drivers explain most of it. Booking too early: 60.5% miss at 31–60 days against 27.8% within a week. Living far: 68.1% miss at 30+ km against 46.5% under 10 km. Having missed before: 55.4% against 43.5%. And the problem is concentrated: patients with 2+ past no-shows hold only 16.4% of appointments but cause 23.2% of all no-shows.

My three recommendations: re-confirm early bookings with a "reply Y" message at two weeks and again 72 hours before; phone-call repeat missers instead of texting them; and offer video visits to far-away Follow-up and General Consultation patients, because 70.6% of far-away no-shows are visit types that can be done remotely.

Why these numbers can be trusted: every headline figure was recomputed by hand in Excel and matched Python exactly; the findings held on a random 20% sample and across both halves of the 18-month period (48.1% vs 48.9%); and the definitions were locked cross-track, so the report, the model and the dashboard all tell one story.

The honest limit: the data is synthetic and shows association, not cause. The clinic should pilot the three actions before a full rollout.

2. Week 7 → Week 8 Transition

1. **Main Week 7 output:** the analytics testing and refinement report with a full test log — KPI cross-checks, chart fixes, filter tests and the cross-track definition check.
2. **Most important testing result:** Python and Excel produced identical KPI values (48.5% overall, 51.2% excluding cancellations). The math is trustworthy.
3. **Major issue discovered:** one chart's Y-axis plotted raw lead days under a percentage title, and an early version used a single colour. A misleading chart is a decision risk.

4. **What was refined:** the chart was rebuilt as percentages with a clear reminded/not-reminded palette; subgroup sizes were added beside every interaction percentage; recommendation wording changed to "pilot first."
5. **Component now ready:** the validated KPI set, the corrected chart pack and the three recommendations — ready for the final dashboard and presentation.
6. **Still to integrate:** the Data Science model's risk score into the repeat-misser phone-call workflow; their final run was still pending at Week 7.
7. **Tracks in final integration:** Data Science (model and definitions) and Project Management (presentation sequence and readiness).
8. **What I will demonstrate:** the final one-page dashboard, the three validated drivers, the cross-track definition lock, and the three recommendations with their numbers.

3. Final Integration Readiness

- **My final component:** a validated analytics evidence base plus a one-page decision-support dashboard (Power BI), backed by a tested Colab notebook.
- **Problem it addresses:** measuring and explaining the no-show problem so the clinic acts on causes, not guesses.
- **Current status:** complete and validated; figures cross-checked in three tools (Python, Excel, Power BI DAX).
- **Improved during Week 7:** misleading chart fixed, subgroup sizes added, all KPIs re-verified by hand and on a sample.
- **Depends on:** Data Science (binary target definition, feature list); Project Management (presentation order, readiness check).
- **Tracks that depend on me:** Data Science (bands and KPI definitions as model features); Project Management (insights and recommendations for the final deck).
- **Output I provide:** cleaned CSV with bands, five KPI definitions with formulas, ranked driver list, final charts, executive summary.
- **Output I need:** Data Science's final model metrics; Project Management's presentation sequence.
- **Evidence of readiness:** Week 7 test log with pass and retest results, Excel-vs-Python match, corrected charts, Data Science questionnaire confirming shared definitions.
- **Limitations that remain:** synthetic data; 90 missing distance and 60 missing waiting-time values; small interaction subgroups; no causal proof.

4. Final KPIs

KPI	FINAL DEFINITION	FINAL VALUE	HOW IT WAS VALIDATED
No-show Rate	no-shows ÷ (attended + no-shows); cancellations excluded, as agreed with Data Science	51.2% (48.5% of all bookings)	Excel pivot matched Python; stable at 48.1% / 48.9% across the two period halves
Reminder Effectiveness	no-show rate reminded vs not, reported per lead-time band	47.4% vs 51.4% overall; ~26 vs 31 (1–7 days), ~40 vs 43.5 (8–30), ~59 vs 63 (30+)	subgroup counts checked; direction consistent in every band
Repeat No show Rate + Concentration	no-show rate for patients with 1+ past miss; share of all no-shows caused by 2+ missers	55.4% vs 43.5%; 16.4% of appointments → 23.2% of no-shows	held on a random 20% sample
Average Booking Lead Days	mean lead days per outcome	34.5 (no-show) vs 24.5 (attended); band gradient 27.8% → 60.5%	matches band charts and Excel
Distance-Band Rate + Telehealth Eligibility	no-show rate per distance band; share of far no-shows in remote-doable types	46.5% → 68.1%; 70.6%	notebook filter test reproduced the cut exactly

5. Validated Findings

1. **The problem is no-shows, not cancellations** — 2,423 against 263. Confirmed in three separate tools.
2. **Lead time is the strongest driver** — 27.8% within a week rising to 60.5% at 31–60 days. Confirmed by band charts, Excel and the sample re-test.
3. **Reminders help a little but cannot fix early bookings** — a reminded 30+ day booking still misses ~59% of the time vs ~63% unreminded. Confirmed with subgroup sizes reported beside each percentage.
4. **Repeat missers are a small, targetable group** — 16.4% of appointments causing 23.2% of no-shows. Confirmed on the 20% sample.
5. **Distance is partly a telehealth problem** — 70.6% of far-away no-shows are Follow-ups or General Consultations. Confirmed by the filter test.
6. **Day, time, type and waiting time are not drivers** (all between 46% and 51%) — validated, and deliberately kept out of the recommendations.

6. Final Dashboard and Key Visualisations

- **One-page Power BI dashboard:** three KPI cards (5,000 appointments; 51.2% no-show rate excluding cancellations; 55.4% repeat-misser rate), two bar charts (no-show by lead-time band and by distance band), and a recommendations panel. Every DAX measure was cross-checked against the Python and Excel values before publishing.
- **Supporting chart pack (in the repository):** outcome mix; lead-time band $\times$ reminder (the corrected chart); repeat-misser concentration; far-away no-show types; plus the Week 5 core charts.
- **Removed or fixed:** the misleading-axis chart was fixed and retested; the waiting-time chart was dropped from the executive view because it carries no decision signal (it stays in the notebook for completeness).

7. Business Insights and Recommendations

1. **Re-confirm early bookings.** A "reply Y" message at two weeks and again 72 hours before, aimed at the 60.5% group. Owner: scheduling team.
2. **Phone-call repeat missers.** The 16.4% of appointments causing 23.2% of the waste deserve a human call, not an SMS. Once available, the Data Science model's risk score can generate this list automatically. Owner: front desk.
3. **Offer video visits to far-away Follow-up and General Consultation patients.** 68.1% miss at 30+ km and 70.6% of those no-shows are remote-doable types. Owner: telehealth pilot.
4. **Keep SMS as the default channel** (45.8%, best of three) but do not expect reminders alone to solve the problem — they move the rate by only ~4 points.
5. **Do not redesign the schedule around day, time or type.** Validated non-drivers; money spent there would not move the number. Each recommendation names an owner, a target group sized by data, and a KPI to watch monthly — with a "normal range" of 48.1–48.9% so the clinic can tell real change from noise.

8. Cross-Track Contribution and Final Integration

1. **Track collaborated with:** Data Science (and Project Management for the final deck).
2. **Dependency:** their model needs my outcome definition and behavioural bands; my recommendation 2 needs their risk score to automate the call list.
3. **Output received:** their Week 6 questionnaire and definition confirmation — binary target (Cancelled dropped), recall-first metric, feature list (`no_show_rate`, `is_weekend`, `is_long_distance`, `appointment_load`), baseline metrics (ROC-AUC 0.628 vs 0.499 dummy).

4. **Output provided:** cleaned CSV with bands, five KPI definitions with formulas, ranked driver list, validated charts, executive summary.
5. **Final integration activity:** a definition lock — one no-show definition across analytics, model and dashboard — plus agreement that their probability score will flag the phone-call list.
6. **What changed:** my KPI 1 formally excludes cancellations; their features encode my distance and history findings; recommendation 2 gained an automation path.
7. **How it improved the solution:** the clinic sees one consistent number on the dashboard, in the model and in the report, instead of three conflicting versions.
8. **Evidence:** shared CSV in the repository, their questionnaire, Week 6 report Section 9, Week 7 test log, GitHub commits.
9. **Contribution to the final presentation:** I own the "what the data says" and "what the clinic should do" sections of the HC-POD walkthrough, and my dashboard is the analytics visual in the final deck.

9. Analytical Limitations

- The data is synthetic and the ~48.5% no-show rate is unrealistically high for a real clinic; treat effect sizes as direction, not exact truth.
- 90 missing distance and 60 missing waiting-time values remain excluded from their own cuts.
- Interaction subgroups are small; sizes are reported beside every percentage.
- Association, not causation — pilot before rollout.
- Single clinic over 18 months; findings may not generalize.
- The automated call-list recommendation waits on Data Science's final model run.